1. Clinical Question & Core Hypothesis

Clinical Question

In HR-positive/HER2-negative breast cancer, high tumor-infiltrating lymphocytes (TILs) are sometimes present yet do not prevent early endocrine therapy failure — why does immune infiltration fail to confer endocrine sensitivity in some cases? 2) Tier 1 paths used: I built on the Ki67 (proliferation) path and the ER/PR IHC path because they provide a mechanistic axis — proliferation (MKI67) and estrogen-receptor signaling (ESR1/ER status) — that modulates therapy responsiveness and interacts with microenvironmental immune signals. 3) Tier 2 evidence that supports the claim: 95GC/95GC-H (signatures linked to targeted therapy response) and circulating microRNA/isomiRs suggest additional, measurable tumor-state dimensions (proliferation/metabolic programs and systemic signaling) that could co-modulate immune function and endocrine sensitivity; CA125 hints at noncanonical systemic signals/inflammation that could also affect immune competence. 4) Why I chose the endpoint 'therapy_response' (endocrine therapy resistance): The clinical problem centers on which HR+/HER2− patients fail endocrine therapy despite apparent favorable features (e.g., immune infiltration); therapy response (endocrine resistance) is the most direct, actionable endpoint to test the proposed mechanism. 5) What I excluded and why: I deliberately excluded Nottingham Prognostic Score from the core anchors to keep the mechanistic hypothesis molecular and cellular (ESR1 + MKI67) and to satisfy the 2-anchor limit; broader prognostic or imaging (TSR, TLS) and exploratory Tier 3 paths were relegated to secondary_modifiers or not used to preserve focus on the immune–proliferation–ER axis?

Working Hypothesis

In HR+/HER2− breast cancer, tumor-infiltrating lymphocytes (TILs) alone do not predict endocrine therapy response; their predictive value is contingent on the tumor's intrinsic proliferation state and estrogen receptor (ER) pathway activity, as well as systemic inflammatory signals. We hypothesize that: (1) high TIL density combined with high proliferation (MKI67) and reduced ER pathway activity (lower ESR1 signaling) predicts endocrine resistance; (2) high TIL density with low proliferation and intact ER signaling predicts endocrine sensitivity; (3) integrating TIL density with MKI67, ESR1 pathway activity, and secondary modifiers (95GC-H signature, circulating miRNAs/isomiRs, CA125) yields a predictive model for endocrine therapy response with superior accuracy over any single biomarker; (4) spatial analysis will show that, in MKI67-high tumors, TILs are more often functionally exhausted and less proximal to tumor cells, explaining diminished therapeutic efficacy; (5) these mechanistic interactions delineate actionable strata for potential combinatorial strategies (e.g., endocrine therapy plus agents that modulate proliferation or reinvigorate T cells) to overcome resistance. Endpoints will include short-term endocrine response (Ki67 reduction after short-term therapy) and long-term outcomes (PFS/DFS).

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MKI67, ESR1
Primary endpoint	therapy_response (endocrine therapy resistance)

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MKI67: MKI67) and to satisfy the 2-anchor limit; broader prognostic or imaging (TSR, TLS) and exploratory Tier 3 paths were relegated to secondary_modifiers or not used to preserve focus on the immune–proliferation–ER axis.

ESR1: ESR1 + MKI67) and to satisfy the 2-anchor limit; broader prognostic or imaging (TSR, TLS) and exploratory Tier 3 paths were relegated to secondary_modifiers or not used to preserve focus on the immune–proliferation–ER axis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	95GC-H	Supporting modifier from knowledge graph
Secondary	95GC	Supporting modifier from knowledge graph
Secondary	circulating microRNA/isomiRs	Supporting modifier from knowledge graph
Secondary	CA125	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: drives tumor dynamics and can reshape ER signaling and the tumor microenvironment, potentially inducing immune exhaustion or altered antigen presentation. ESR1 pathway activity defines ER-driven transcriptional programs essential for endocrine sensitivity.

Prediction 2: capture systemic/intrinsic tumor states that can modulate immune function and treatment response. By integrating local (TILs, MKI67, ESR1 activity) with systemic (95GC-H, miRNA/isomiR, CA

Prediction 3: information, we aim to build a robust, mechanistically grounded predictor of endocrine response that can guide personalized therapy decisions. The proposed approach aligns with current understanding of tumor–immune–proliferation crosstalk and leverages feasible assays (IHC, multiplex IF, RNA sequencing, and targeted circulating biomarker assays) in retrospective and prospective cohorts.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- 95GC-H: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- 95GC: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.